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Group 1 MATH 341

Homework 6

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Problem 1

The objective of this problem is to approximate the integral $I(f) = \int_0^{\pi} e^x \cos(x) dx$ using

Gauss-Legendre quadrature for $n = 2$ and $n = 3$. That is, using $\int_{-1}^1 f(x) dx \approx \sum_{j=1}^n w_j f(x_j)$. The true value is known to be approximately -12.07034631638963. Naturally, a change of variables must take place so that the integral is bounded from -1 to 1. This working is shown below,

HW 6 #1
 $I(f) = \int_0^{\pi} e^x \cos(x) dx$ True value: $\frac{e^{\pi+1}-1}{2} \approx -12.07034631638963$
change variables $0 < x < \pi$ to $-1 < u < 1$
 $0 < \frac{2x}{\pi} < 2$ $0 < u+1 < 2$
 $\frac{2x}{\pi} = u+1 \rightarrow u = \frac{2x}{\pi} - 1$ then $x = \frac{\pi}{2}(u+1)$ $dx = \frac{\pi}{2} du$ substitute in
So $I(f) = \int_0^{\pi} e^x \cos(x) dx = \frac{\pi}{2} \int_{-1}^1 e^{\frac{\pi}{2}(u+1)} \cos(\frac{\pi}{2}(u+1)) du$
 $I(f) = \frac{\pi}{2} \int_{-1}^1 e^{\frac{\pi}{2}(u+1)} \cos(\frac{\pi}{2}(u+1)) du \approx \sum_{j=1}^n w_j f(u_j)$

To compute the approximations more cleanly, we used Octave coding. Below is the code used for $n = 2$, using values for the weights and nodes from Table 5.10 in Atkinson:

```
1| format long;  
2| fhandle1 = @(x)((pi/2)*(exp((pi/2)*(x+1)).*cos(pi/2*(x+1))));  
3| x1 = -0.5773502692; x2 = -x1;  
4| w1 = 1; w2 = w1;  
5| w1*fhandle1(x1)+w2*fhandle1(x2)  
Output: - 12.33621046613186
```

The following code computed the approximation for $n = 3$, once again using Table 5.10:

```
1| format long;  
2| fhandle = @(x)((pi/2)*(exp((pi/2)*(x+1)).*cos(pi/2*(x+1))));  
3| x1 = -0.7745966692; x2 = 0; x3 = -x1;  
4| w1 = 0.5555555556; w2 = 0.8888888889; w3 = w1;  
5| w1*fhandle(x1)+w2*fhandle(x2)+w3*fhandle(x3)  
Output: - 12.12742044991054
```

As seen, the approximation for $n = 3$ is closer to the true value. Precise errors were calculated by-hand in the image below. It is clear that the use of more nodes leads to a more accurate approximation.

$$\begin{aligned}\underline{\text{Error when } n=2} &= \left| -12.07034627 - (-12.33619904) \right| \\ &= \boxed{0.26585277}\end{aligned}$$

$$\begin{aligned}\underline{\text{Error when } n=3} &= \left| -12.07034627 - (-12.12743991) \right| \\ &= \boxed{0.057093640}\end{aligned}$$

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Problem 2

The conditions for the degree of precision m of a quadrature are as follows:

- 1) $I(f) = I_n(f)$ for polynomials of degree $0 \leq k \leq m$, exact for $f(x) = x^k$, $0 \leq k \leq m$
- 2) $I(f) \neq I_n(f)$ for some polynomial of degree $m + 1$, integration *not* exact for $f(x) = x^{m+1}$

(2)

Find constants x_1, x_2 , and w_1, w_2 such that quadrature formula

$$\int_0^1 f(x) dx \approx \frac{1}{2} f(x_1) + w_2 f(x_2)$$

has highest possible degree of precision m .

know: $n=2$, $w_1 = \frac{1}{2}$ integration is exact for $f(x) = x^k$, $0 \leq k \leq m$, not for $f(x) = x^{m+1}$

→ highest possible = 3, so $k = 0, 1, 2$

we have polynomials

$$f_0(x) = x^0 = 1 \rightarrow \int_0^1 f_0(x) dx = 1 = \frac{1}{2} f_0(x_1) + w_2 f_0(x_2) = \frac{1}{2} + w_2$$

$$f_1(x) = x^1 = x \rightarrow \int_0^1 f_1(x) dx = \frac{1}{2} = \frac{1}{2} f_1(x_1) + w_2 f_1(x_2) = \frac{1}{2} x_1 + w_2 x_2$$

$$f_2(x) = x^2 \rightarrow \int_0^1 f_2(x) dx = \frac{1}{3} = \frac{1}{2} f_2(x_1) + w_2 f_2(x_2) = \frac{1}{2} x_1^2 + w_2 x_2^2$$

then we have the system of non-linear equations

$$\begin{aligned} \frac{1}{2} + w_2 &= 1 \\ \frac{1}{2} x_1 + w_2 x_2 &= \frac{1}{2} \\ \frac{1}{2} x_1^2 + w_2 x_2^2 &= \frac{1}{3} \end{aligned}$$

w_2 is clearly $\frac{1}{2}$, then we have the two equations (with two unknowns)

$$\begin{aligned} \frac{1}{2} x_1 + \frac{1}{2} x_2 &= \frac{1}{2} \\ \frac{1}{2} x_1^2 + \frac{1}{2} x_2^2 &= \frac{1}{3} \end{aligned}$$

rearrange, so $x_1 = 1 - x_2$, then

$$\begin{aligned} \frac{1}{2} (1 - x_2)^2 + \frac{1}{2} x_2^2 &= \frac{1}{3} \\ \frac{1}{2} (1 - 2x_2 + x_2^2) + \frac{1}{2} x_2^2 &= \frac{1}{3} \\ \frac{1}{2} - x_2 + x_2^2 &= \frac{1}{3} \\ \Rightarrow x_2^2 - x_2 + \frac{1}{6} &= 0 \end{aligned}$$

using quadratic formula: $x_2 = \frac{1 \pm \sqrt{1 - \frac{4}{6}}}{2} = \frac{1 \pm \sqrt{\frac{1}{3}}}{2}$ or approx. 0.211 and 0.789

$x_1 = \frac{1 - \sqrt{1/3}}{2}$ $x_2 = \frac{1 + \sqrt{1/3}}{2}$

Choose $x_2 = 0.789$, so that $x_1 = 1 - 0.789 \approx 0.211$ and $x_1 < x_2$

assembling all information, we have $I_2(f) = \frac{1}{2} f(0.211) + \frac{1}{2} f(0.789)$ or more precisely $I_2(f) = \frac{1}{2} f(\frac{1-\sqrt{1/3}}{2}) + \frac{1}{2} f(\frac{1+\sqrt{1/3}}{2})$

with degree of precision $m=3$

Show $I(f) \neq I_n(f)$, i.e. for $f(x) = x^{m+1}$, or for $f(x) = x^4$

we have $f_4(x) = x^4 \rightarrow \int_0^1 f_4(x) dx = \frac{1}{5} = \frac{1}{2} f_4(x_1) + \frac{1}{2} f_4(x_2)$

check: $\frac{1}{2} (0.211)^4 + \frac{1}{2} (0.789)^4 \approx \frac{1}{5}$ ✓

we have $f_4(x) = x^4 \rightarrow \int_0^1 f_4(x) dx = \frac{1}{5} = \frac{1}{2} x_1^4 + \frac{1}{2} x_2^4$

check: $\frac{1}{2} (0.211)^4 + \frac{1}{2} (0.789)^4 \approx 0.194 \neq \frac{1}{5}$

The values for the nodes x_1 and in turn x_2 were calculated using the Octave code:

- 1| format long;
- 2| x1 = (1-sqrt(1/3))/2
- 3| x2 = (1+sqrt(1/3))/2

Resulting in precise values $x_1 = 0.211324865405187$ and $x_2 = 0.788675134594813$. These

were rounded to three decimal places for readability in the written portion of the problem. It can be seen that the approximation $I_n(f)$ is exact for k up to $m = 3$, but not for $m + 1 = 4$. Also

worth mentioning is the non-uniqueness of the solution because the quadratic formula was used. This shows the idea that the uniqueness of the solution is up to the ordering of nodes.

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Problem 3

The objective here is to derive the two-point Gaussian quadrature formula. Since $n = 2$, and the degree of precision is known to be $2n - 1$, then $m = 3$. The derivation of the equations is shown below following the quadrature conditions for degree of precision m .

The conditions are the same as the previous problem:

- 3) $I(f) = I_n(f)$ for polynomials of degree $0 \leq k \leq m$, exact for $f(x) = x^k$, $0 \leq k \leq m$
- 4) $I(f) \neq I_n(f)$ for some polynomial of degree $m + 1$, integration *not* exact for $f(x) = x^{m+1}$

3) $I(f) = \int_a^b w(x)f(x)dx$ with weight function $w(x) = \ln(\frac{1}{x}) = -\ln(x)$ on interval $(a,b) = (0,1)$

Derive two-point Gaussian quadrature formula ($n=2$), finding nodes x_1, x_2 , and weights w_1, w_2 .

State results $I_2(f) = w_1 f(x_1) + w_2 f(x_2)$ with $x_1 < x_2$

degrees of precision: $2n-1 \rightarrow m=3$ so $f_i(x) = x^i$, $i=0,1,2,3$, $i=1,2,3,4$

$f_0(x) = x^0 = 1 \rightarrow -\int_0^1 \ln x \cdot 1 dx = 1 = w_1 f_0(x_1) + w_2 f_0(x_2) = w_1 + w_2$
 $f_1(x) = x^1 = x \rightarrow -\int_0^1 x \ln x dx = \frac{1}{4} = w_1 f_1(x_1) + w_2 f_1(x_2) = w_1 x_1 + w_2 x_2$
 $f_2(x) = x^2 \rightarrow -\int_0^1 x^2 \ln x dx = \frac{1}{9} = w_1 f_2(x_1) + w_2 f_2(x_2) = w_1 x_1^2 + w_2 x_2^2$
 $f_3(x) = x^3 \rightarrow -\int_0^1 x^3 \ln x dx = \frac{1}{16} = w_1 f_3(x_1) + w_2 f_3(x_2) = w_1 x_1^3 + w_2 x_2^3$

re-written below

$\int_0^1 \ln x dx$	$\int_0^1 x \ln x dx$	$\int_0^1 x^2 \ln x dx$	$\int_0^1 x^3 \ln x dx$
$u = \ln x \quad v = x$	$u = \ln x \quad v = x^2/2$	$u = \ln x \quad v = x^3/3$	$u = \ln x \quad v = x^4/4$
$du = \frac{1}{x} dx \quad dv = 1$	$du = \frac{1}{x} dx \quad dv = x$	$du = \frac{1}{x} dx \quad dv = x^2$	$du = \frac{1}{x} dx \quad dv = x^3$
$= (x \ln x)' - \int_0^1 1 dx = 1$	$= -\left(\frac{x^2}{2} \ln x\right)' - \int_0^1 \frac{x}{2} dx = \frac{1}{4}$	$= -\left(\frac{x^3}{3} \ln x\right)' - \int_0^1 \frac{x^2}{3} dx = \frac{1}{9}$	$= -\left(\frac{x^4}{4} \ln x\right)' - \int_0^1 \frac{x^3}{4} dx = \frac{1}{16}$

We have the equations

$$\begin{cases} w_1 + w_2 = 1 \\ w_1 x_1 + w_2 x_2 = \frac{1}{4} \\ w_1 x_1^2 + w_2 x_2^2 = \frac{1}{9} \\ w_1 x_1^3 + w_2 x_2^3 = \frac{1}{16} \end{cases}$$

Solve: $\rightarrow$ by hand
 $\rightarrow$ Octave

To solve the system of nonlinear equations, we used the following Octave code after loading the corresponding package with “pkg load symbolic”.

```

1| format long;
2| syms x1 x2 w1 w2;
3| f1 = w1 + w2 - 1;
4| f2 = w1*x1+w2*x2-1/4;
5| f3 = w1*x1^2+w2*x2^2-1/9;
6| f4 = w1*x1^3+w2*x2^3-1/16;
7| S = solve(f1,f2,f3,f4)

```

```
>> S{2}.w1
ans = (sym)
```

$$\frac{9\sqrt{106}}{424} + \frac{1}{2}$$

```
>> S{2}.w2
ans = (sym)
```

$$\frac{1}{2} - \frac{9\sqrt{106}}{424}$$

```
>> S{2}.x1
ans = (sym)
```

$$\frac{5}{14} - \frac{\sqrt{106}}{42}$$

```
>> S{2}.x2
ans = (sym)
```

$$\frac{\sqrt{106}}{42} + \frac{5}{14}$$

The image to the left is the technique used to access the solutions calculated using the solve() function in Octave. For readability, the results are shown below in both exact form and to 10 decimal places.

$$w_1 = \frac{1}{2} + \frac{9\sqrt{106}}{424} \approx 0.7185393190$$

$$w_2 = \frac{1}{2} - \frac{9\sqrt{106}}{424} \approx 0.2814606809$$

$$x_1 = \frac{5}{14} - \frac{\sqrt{106}}{42} \approx 0.1120088061$$

$$x_2 = \frac{5}{14} + \frac{\sqrt{106}}{42} \approx 0.6022769081$$

Checking these values in the equations, it can be seen that

$$w_1 + w_2 = 1,$$

$$w_1 x_1 + w_2 x_2 = 1/4,$$

$$w_1 x_1^2 + w_2 x_2^2 = 1/9,$$

$$w_1 x_1^3 + w_2 x_2^3 = 1/16, \text{ and}$$

$$w_1 x_1^4 + w_2 x_2^4 \approx 0.0371472663 \neq 1/25, \text{ therefore the conditions have}$$

been met. It should also be noted that the code produced two separate solutions, meaning there is not a unique solution to the equations.

Specifically, the first solution is identical but with the inner signs changed (that is, w_1 and w_2 are switched, and x_1 and x_2 are switched).

The second solution was selected for presentation as it presented the nodes in ascending order. This could be due to the solution involving the use of the quadratic formula, but also due to the fact that $n = 2$ and the nodes (and thereby weights) are expected to be symmetrical

about the center of the integration from 0 to 1.

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